

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
NMTC JUNIOR LEVEL – IX & X Grades FINAL

NMTC – 19th OCTOBER, 2024

Instructions:

1. Answer all questions. Each question carries 10 marks.
 2. Elegant and innovative solutions will get extra marks.
 3. Diagram and Justification should be given wherever necessary.
 4. Before start answering, fill in the FACE SLIP completely.
 5. Your 'rough work' should be in the answer sheet itself.
 6. Maximum Time allowed is THREE hours.
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1. Find the number of sets of 4 positive integers less than or equal to 25, such that the difference between any two elements in the set is at least 3.
2. Solve the system of equations:
$$(1 + 4^{2x-y})5^{1-2x+y} = 1 + 2^{2x-y+1}$$
$$y^3 + 4x + 1 + \log(y^2 + 2x) = 0$$
3. a, b, c, d, e, f are real numbers such that the polynomial
$$p(x) = x^8 - 4x^7 + 7x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$$
factorises into eight linear factors $(x - x_i)$ with $x_i > 0$, for $i = 1, 2, 3, \dots, 8$. Find the possible values of the constant f .
4. An acute angled triangle PQR is inscribed in a circle. Given $\angle P = 60^\circ$, $\angle R > \angle Q$. Let H and I be the orthocenter and incenter of ΔPQR . Find the ratio of $\angle PHI$ to $\angle PQR$.
5. ABC is a triangle in which $\angle A$ is obtuse. AD, BE and CF are the altitudes from A, B and C respectively to BC, CA and AB . A_1, B_1 and C_1 are arbitrary points on BC, CA and AB respectively. The circle on AA_1, BB_1 and CC_1 as diameters are drawn. Show that the lengths of the tangents drawn from the orthocenter of ΔABC to these circles are equal.
6. Find all-natural number solutions of the equation $3^x - 5^y = z^2$.
7. a, b, c are positive reals such that $ab + bc + ca = 3abc$. Show that $a^2b + b^2c + c^2a \geq 2(a + b + c) - 3$. When will equality hold?
8. A regular polygon with 100 vertices is given. To each vertex, a natural number from the set $\{1, 2, 3, \dots, 49\}$ is assigned. Prove that there are four vertices A, B, C, D such that if the numbers a, b, c, d are assigned to them respectively, then $a + b = c + d$ and $ABCD$ is a parallelogram.
